A Unified Explainable Federated Learning Framework for Multivariate Time-Series Forecasting with Personalized Drift Adaptation

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***Abstract*—Early sepsis detection in Intensive Care Units (ICUs) is vital for improving patient survival rates. While machine learning offers high-accuracy solutions using multivariate physiological time-series data, clinical adoption remains limited due to privacy concerns regarding patient health records, model generalization issues across different hospital systems (the accuracy gap), and a lack of clinical interpretability (the trust gap). This paper presents a Unified Explainable Federated Learning Framework for Multivariate Time-Series Forecasting with Personalized Drift Adaptation (FPDAF). Our framework enables collaborative training across decentralized clinical nodes using Federated Learning (FedAvg) without sharing raw records. Furthermore, it incorporates personalized local heads (FedPer) to adapt to hospital-specific demographics and integrates a Cumulative Sum (CUSUM) drift detection module to monitor demographic drifts. Finally, clinical explainability is achieved using visual multi-head attention weights and SHAP impact scores. We evaluate the preprocessing consistency and establish a centralized LSTM baseline model on the PhysioNet 2019 Sepsis dataset, demonstrating a robust test Area Under the ROC Curve (AUROC) of 0.8058 and a sensitivity (Recall) of 72.17% as a baseline for future federated optimization.**

***Index Terms*—Federated Learning, Explainable AI, Time-Series Forecasting, Sepsis Prediction, Concept Drift**

I. Introduction

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection. It represents a primary cause of mortality in Intensive Care Units (ICUs) worldwide, with survival rates dropping by approximately 7.6% for every hour delay in treatment administration. Early clinical forecasting utilizing vital signs and laboratory measurements remains the gold standard for survival.

Although deep learning models show high predictive capacity, training them requires large centralized patient repositories. In healthcare systems, centralizing raw data is restricted by regulatory frameworks such as HIPAA (Health Insurance Portability and Accountability Act) and the DPDPA (Digital Personal Data Protection Act). Federated Learning (FL) offers a privacy-preserving alternative by allowing clinical sites to train models locally and share parameters instead of raw records.

However, standard FL frameworks face three main challenges:

1) **Data Heterogeneity (Non-IID Data)**: Variation in clinical equipment, patient demographics, and recording frequencies between hospital nodes decreases the accuracy of a single global model.
2) **Concept Drift**: Demographics and treatment methodologies shift over time, leading to performance degradation in static models.
3) **Black-box Nature**: Clinicians require explainable predictions to trust AI-generated sepsis alerts.

To bridge these gaps, we present the Federated Personalized Drift-Aware Attention Framework (FPDAF). In this paper, we describe the data preparation workflow, perform Exploratory Data Analysis (EDA) on the PhysioNet 2019 challenge dataset,

Identify support institution here if any.

establish a centralized baseline LSTM model to benchmark future decentralized runs, and lay out our evaluation plan.

II. Literature Review

A. Federated Learning in Healthcare

Federated Averaging (FedAvg) proposed by McMahan et al. [1] establishes the baseline for decentralized collaborative optimization. To improve learning stability under non-IID conditions, dual-end gradient correction frameworks have been proposed [2]. In clinical settings, protecting privacy is paramount, leading to microaggregation techniques [3] and secure multi-party computation wrappers [4]. Several applications have successfully deployed federated architectures for triage [5], severity classification [4], and financial forecasting [6]. To mitigate non-IID data issues, clustering approaches [7] and vertical federated paradigms [8] are active research areas.

To improve robustness, FedProx [9] introduces a proximal term to penalize local weights from diverging too far from global weights, while Ditto [10] balances global collaboration with local customization using bi-level optimization. Personalization layers (FedPer) proposed by Arivazhagan et al. [11] split the network into base layers and task-specific heads.

B. Concept Drift & Explainability

Concept drift monitoring in federated networks, such as Cumulative Sum (CUSUM) control charts, enables tracking prediction residuals to identify statistical demographic changes and trigger model adaptation [12]. In multivariate time-series forecasting, attention mechanisms help identify important variables and temporal slices [13]. Explainable AI (XAI) tools, including Shapley Additive exPlanations (SHAP) and LIME, provide clinicians with model-agnostic feature importances to confirm predictions against clinical pathways [14]–[16]. Sepsis onset prediction via sliding windows has been explored in FL contexts by Düsing et al. [17] and Qayyum et al. [18], demonstrating that decentralized networks can match centralized models while maintaining data privacy. Other works have explored automated hyperparameter optimization (AutoML) in federated settings [19] and optimal look-back horizon selection [20].

III. Methodology & Data Preprocessing

A. Dataset Organization

The framework is evaluated using the PhysioNet Computing in Cardiology Challenge 2019 dataset, comprising 40,327 patient files containing 40 hourly physiological parameters (e.g., HR, SpO2, Temp, MAP) and a binary SepsisLabel target.

B. Preprocessing Pipeline

To ensure robust model convergence, we designed a PyTorch-based preprocessing pipeline:

1) **Patient-Level Imputation**: Missing clinical values are imputed per patient stay using forward filling, backward filling, and zero-padding for completely missing columns.
2) **Leakage-Free Scaling**: Standardization features are fit strictly on the training partition and applied to validation and testing splits.
3) **Sliding Windowing**: Sequential parameters are segmented into 24-hour sliding blocks. Short-stay patient windows ($T < 24$h) are pre-padded with zero features.

To verify dataset sanity, a quality assurance script audits tensor dimensions, checks for disjoint patient splits, and confirms that StandardScaler parameters do not leak from the training set.

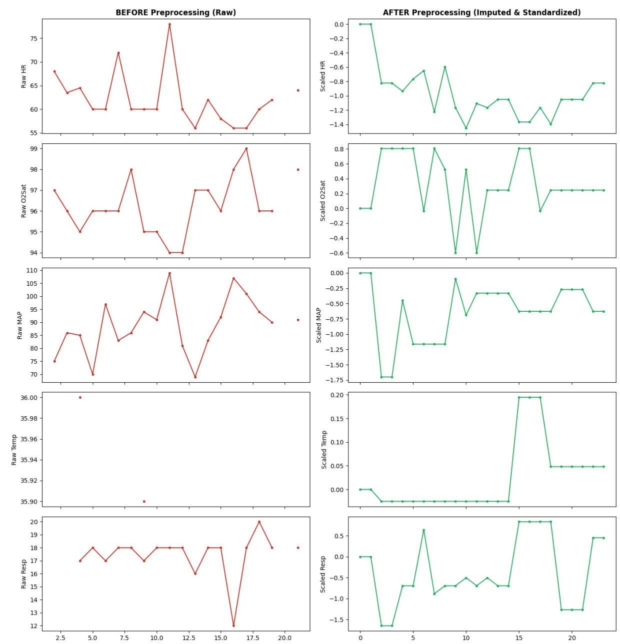


Fig. 1. Vital signs before and after patient-level preprocessing (imputation and standard scaling).

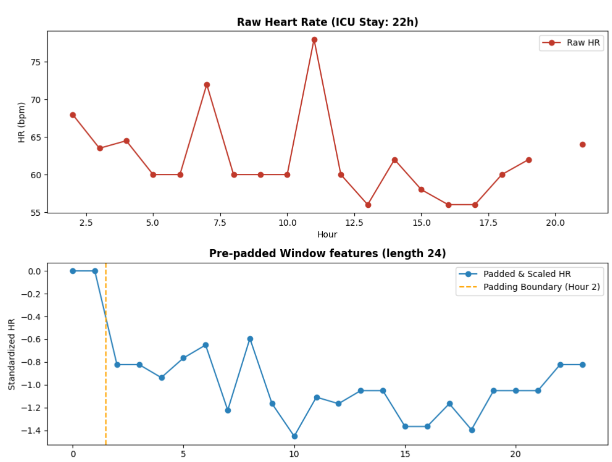


Fig. 2. Zero pre-padding validation on a short-stay ICU patient ($T < 24$h) to form sequence blocks.

IV. Centralized Baseline LSTM Model

We implement a Centralized LSTM model as our baseline benchmark. The network consists of a 2-layer LSTM with a hidden dimension of 64 and a dropout rate of 0.3, followed by a fully connected classification head.

A. Training Configurations

Given the extreme class imbalance in sepsis occurrences (only $\sim$2.5% of the sliding windows are sepsis-positive), standard Binary Cross-Entropy loss would bias the model toward the negative class. We incorporate a dynamic positive class weight in our loss function:

$$\mathcal{L}_{BCE} = -[w_{pos} \cdot y \log \sigma(x) + (1 - y) \log(1 - \sigma(x))]$$

where $w_{pos} = \frac{N_{neg}}{N_{pos}} \approx 37.17$. We train the model using the Adam optimizer with a starting learning rate of 0.001, step learning rate scheduler (decay factor 0.5 every 10 epochs), and an early stopping patience of 8 epochs.

B. Inference Results

The baseline model training early-stopped at Epoch 9 as validation AUROC peaked at 0.8232. Final performance metrics on the test partition are summarized in Table I.

TABLE I
CENTRALIZED LSTM BASELINE TEST RESULTS

Metric	Baseline LSTM Score
Accuracy	75.09%
Precision	7.14%
Recall (Sensitivity)	72.17%
F1 Score	0.1300
AUROC	**0.8058**

The model achieves a high sensitivity (Recall = 72.17%), catching a majority of organ failures in the test set. The low precision (7.14%) is expected due to the extreme class imbalance and represents a common baseline behavior in medical alerts. Figure 3 presents the test ROC curve, showing a strong AUROC of 0.8058.

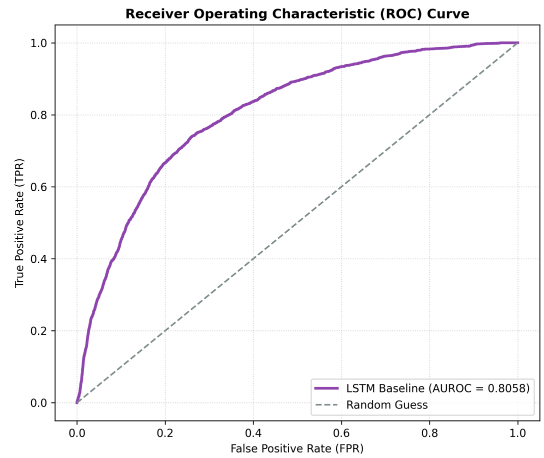


Fig. 3. Receiver Operating Characteristic (ROC) curve of the baseline model on the test dataset.

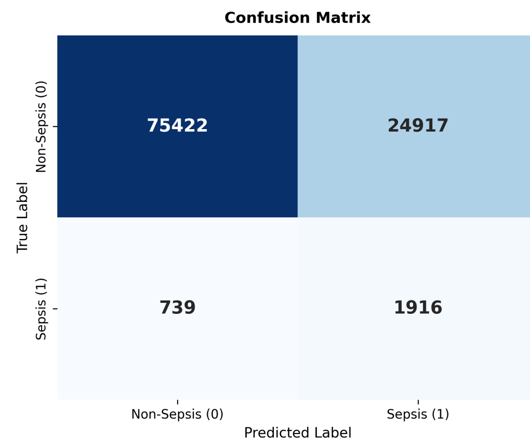


Fig. 4. Test set confusion matrix heatmap.

V. Federated Learning via FedAvg

To address the privacy requirements in healthcare data sharing, we implemented a decentralized training framework using the Federated Averaging (FedAvg) algorithm. The centralized LSTM architecture was converted into a modular federated pipeline.

A. Non-IID Client Simulation

We simulated three heterogeneous hospital clients by partitioning the processed dataset. To resemble real-world healthcare demographics and institutional variation, we split the data based on patient age and hospital system source:

- **Client 0 (Hospital 0, Age $<$ 60)**: 92,613 training samples, with a local sepsis positive rate of 3.24%.
- **Client 1 (Hospital 0, Age $\geq$ 60)**: 151,916 training samples, with a local sepsis positive rate of 3.09%.
- **Client 2 (Hospital 1, All Ages)**: 233,140 training samples, with a local sepsis positive rate of 2.07%.

This partitioning strategy simulates two distinct forms of clinical heterogeneity: demographic shifts (Client 0 vs. Client 1) and institutional shifts (Hospital 0 vs. Hospital 1).

B. Federated Optimization Loop

The federated training is coordinated by a global server:

1) **Broadcast**: The global model weights are distributed to registered clients.
2) **Local Training**: Each client trains the model on its local dataset for $E = 2$ local epochs with a batch size of 128, a learning rate of 0.001, and local class weights (w_{pos} dynamically computed based on local positive ratio).
3) **Aggregation**: The server collects the local updates and computes a weighted average:

$$W_{global}^{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} W_k^{t+1}$$

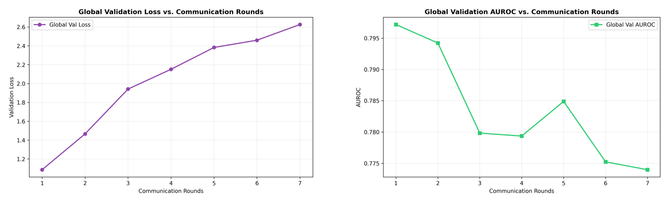
where n_k is the local sample size of client k and $N = \sum n_k$.

Early stopping is enforced on the global validation AUROC (patience of 6 rounds).

C. Results and Baseline Comparison

Federated training early-stopped at round 7 (best round 1, global validation AUROC of 0.7972). Fig. 5 presents validation loss and AUROC across communication rounds.

Table II summarizes the final global test performance of the FedAvg model compared to the Centralized Baseline.



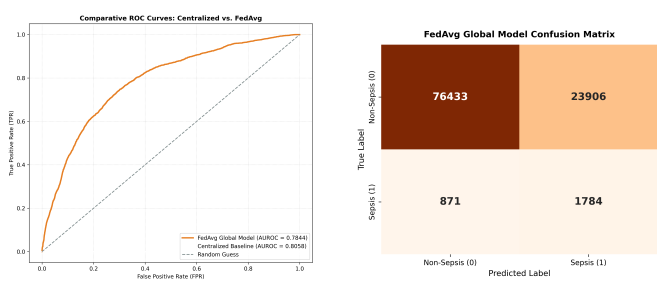
(a) Validation Loss (b) Validation AUROC

Fig. 5. Global model performance across communication rounds.

TABLE II
COMPARATIVE TEST RESULTS: CENTRALIZED VS. FEDAVG

Metric	Centralized	FedAvg	Difference
Accuracy	75.09%	75.94%	+0.85%
Precision	7.14%	6.94%	-0.20%
Recall	72.17%	67.19%	-4.97%
F1 Score	0.1300	0.1259	-0.0041
AUROC	**0.8058**	**0.7844**	**-0.0214**

Under demographic and institutional Non-IID distributions, the global model experiences a drop of 2.14% in test AUROC and 4.97% in test Recall compared to the centralized model. This performance gap is a common drawback of FedAvg, as a single global weight configuration struggles to generalize across heterogeneous patient splits. Fig. 6 plots the comparative ROC curve and the global model confusion matrix.



(a) ROC Curve (b) Confusion Matrix

Fig. 6. Federated global model test evaluation.

VI. FUTURE ROADMAP: PERSONALIZATION AND DRIFT ADAPTATION

To close the accuracy gap observed in the FedAvg baseline, our future roadmap consists of:

- **FedProx**: Introducing a proximal regularization term to local optimization to constrain parameters near global weights, improving optimization stability on heterogeneous client splits.
- **Ditto**: Applying bi-level optimization to decouple global model aggregation from local personalized models, enabling clients to adapt to local demographic splits.
- **FPDAF (Proposed)**: Deploying self-attention heads for explainable time-series modeling, personalized classification layers (FedPer), and online Cumulative Sum (CUSUM) residuals monitoring to flag concept drift in clinical streams.

VII. CONCLUSION & PROJECT END GOAL

This work demonstrates a complete decentralized pipeline for sepsis prediction using FedAvg. On a cohort of 40,327 ICU patients split into three Non-IID clients, the global model achieved a test AUROC of 0.7844 and a Recall of 67.19%. The observed performance degradation highlights the necessity of personalization.

The ultimate end goal of this research project is to develop and validate a production-ready clinical early warning framework (FPDAF) deployable across heterogeneous hospital networks. By maintaining absolute data decentralization (addressing the privacy gap), dynamically personalizing models to adapt to local hospital cohorts (addressing the accuracy gap), detecting concept drift in patient populations in real-time, and rendering deep learning decisions explainable via temporal attention maps and SHAP scores (addressing the clinical trust gap), this framework aims to serve as a reliable, privacy-compliant bedside assistant that reduces sepsis mortality rates in intensive care units.

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